

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): D20WU-HS, D20HD-HS, D16HD-HS, D16WU-HS

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.02.0000-b013	2.2.0

Version History

Module Version	Date	Notes
1_0_1_0	4/25/2019	Updated communication sheet with notes for configuring HDBaseT settings. Fixed command string for PIP Input command.
1_0_0_0	4/12/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='D20WU-HS')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='D20WU-HS')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='D20WU-HS')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' 'LetterBox' 'Full Height'	Value 'Native' 'Full Size' 'Custom'	Value '4:3' 'Full Width' '3D Mode'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AutoImage ¹	Value 'Normal'	Value 'Wide'	
# AutoImage example InterfaceName.Set('AutoImage', 'Normal')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'VGA' 'DVI-D'	Value 'HDMI 1' '3G-SDI'	Value 'HDMI 2' 'HDBaseT'
# Input example InterfaceName.Set('Input', 'VGA')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command LampMode	Value 'Constant Power' 'ECO 2'	Value 'Constant Intensity' 'Rental Mode'	Value 'ECO 1'
# LampMode example InterfaceName.Set('LampMode', 'Constant Power')			
Command MenuNavigation	Value 'Up' 'Right' 'Menu'	Value 'Down' 'Enter'	Value 'Left' 'Exit'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')			
Command PIP	Value 'On'	Value 'Off'	
# PIP example InterfaceName.Set('PIP', 'On')			
Command PIPIInput	Value 'VGA' 'DVI-D'	Value 'HDMI 1' '3G-SDI'	Value 'HDMI 2' 'HDBaseT'

# PIPInput example InterfaceName.Set('PIPInput', 'VGA')			
Command PIPPosition	Value 'Left Vertical Center' 'Bottom Center' 'Top Left'	Value 'Top Center' 'Bottom Right' 'Top Right'	Value 'Right Vertical Center' 'Bottom Left'
# PIPPosition example InterfaceName.Set('PIPPosition', 'Left Vertical Center')			
Command PIPSize	Value 'Small'	Value 'Medium'	Value 'Large'
# PIPSize example InterfaceName.Set('PIPSize', 'Small')			
Command PIPSwap	Value 'None'		
# PIPSwap example InterfaceName.Set('PIPSwap', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			

¹ Normal Mode supports all 4:3 input sources. Wide Mode supports all 16:9 input sources and most 4:3 sources.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Auto' 'LetterBox' 'Full Height'	Value 'Native' 'Full Size' 'Custom'	Value '4:3' 'Full Width' '3D Mode'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AutoImage	Value 'Normal'	Value 'Wide'	
# AutoImage example InterfaceName.Update('AutoImage') Value = InterfaceName.ReadStatus('AutoImage') InterfaceName.SubscribeStatus('AutoImage', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'VGA' 'DVI-D'	Value 'HDMI 1' '3G-SDI'	Value 'HDMI 2' 'HDBaseT'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Constant Power' 'ECO 2'	Value 'Constant Intensity' 'Rental Mode'	Value 'ECO 1'
# LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command OnScreenDisplay	Value 'On'	Value 'Off'	

<pre># OnScreenDisplay example InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)</pre>			
Command PIP	Value 'On'	Value 'Off'	
<pre># PIP example InterfaceName.Update('PIP') Value = InterfaceName.ReadStatus('PIP') InterfaceName.SubscribeStatus('PIP', None, FeedbackHandler)</pre>			
Command PIPInput	Value 'VGA' 'DVI-D'	Value 'HDMI 1' '3G-SDI'	Value 'HDMI 2' 'HDBaseT'
<pre># PIPInput example InterfaceName.Update('PIPInput') Value = InterfaceName.ReadStatus('PIPInput') InterfaceName.SubscribeStatus('PIPInput', None, FeedbackHandler)</pre>			
Command PIPPosition	Value 'Left Vertical Center' 'Bottom Center' 'Top Left'	Value 'Top Center' 'Bottom Right' 'Top Right'	Value 'Right Vertical Center' 'Bottom Left'
<pre># PIPPosition example InterfaceName.Update('PIPPosition') Value = InterfaceName.ReadStatus('PIPPosition') InterfaceName.SubscribeStatus('PIPPosition', None, FeedbackHandler)</pre>			
Command PIPSize	Value 'Small'	Value 'Medium'	Value 'Large'
<pre># PIPSize example InterfaceName.Update('PIPSize') Value = InterfaceName.ReadStatus('PIPSize') InterfaceName.SubscribeStatus('PIPSize', None, FeedbackHandler)</pre>			
Command Power	Value 'On'	Value 'Off'	
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Refer to the following notes regarding setting the serial port path:

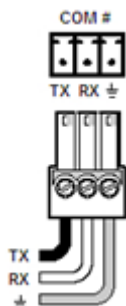
- Set the serial port path to allow the command pass through RS232 or HDBaseT.
 1. Select Menu > Configuration > Communications > Serial Port Path.
 2. Select the appropriate serial port path.
 3. To confirm the selection, press ENTER.

Serial communication

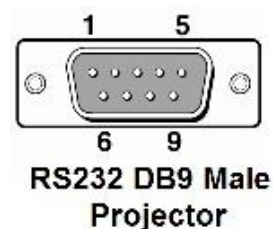
Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface.

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 3D Mode	(SZP 8)
Aspect Ratio 4:3	(SZP 2)
Aspect Ratio Auto	(SZP 0)
Aspect Ratio Custom	(SZP 7)
Aspect Ratio Full Height	(SZP 6)
Aspect Ratio Full Size	(SZP 4)
Aspect Ratio Full Width	(SZP 5)
Aspect Ratio LetterBox	(SZP 3)
Aspect Ratio Native	(SZP 1)
Auto Image Normal	(AIG 0)
Auto Image Wide	(AIG 1)
Freeze Off	(FRZ 0)
Freeze On	(FRZ 1)
Input 3G-SDI	(SIN+MAIN 7)
Input DVI-D	(SIN+MAIN 5)
Input HDBaseT	(SIN+MAIN 8)
Input HDMI 1	(SIN+MAIN 3)
Input HDMI 2	(SIN+MAIN 4)
Input VGA	(SIN+MAIN 1)
Keypad 0	(KEY 36)
Keypad 9	(KEY 34)
Lamp Mode Constant Intensity	(LPM 1)
Lamp Mode Constant Power	(LPM 0)
Lamp Mode ECO 1	(LPM 2)
Lamp Mode ECO 2	(LPM 3)
Lamp Mode Rental Mode	(LPM 4)
Menu Navigation Down	(KEY 42)
Menu Navigation Enter	(KEY 40)
Menu Navigation Exit	(KEY 20)
Menu Navigation Left	(KEY 39)
Menu Navigation Menu	(KEY 19)
Menu Navigation Right	(KEY 41)
Menu Navigation Up	(KEY 38)
On Screen Display Off	(OSD 0)
On Screen Display On	(OSD 1)
PIP Input 3G-SDI	(SIN+PIIP 7)
PIP Input DVI-D	(SIN+PIIP 5)
PIP Input HDBaseT	(SIN+PIIP 8)
PIP Input HDMI 1	(SIN+PIIP 3)
PIP Input HDMI 2	(SIN+PIIP 4)
PIP Input VGA	(SIN+PIIP 1)
PIP Off	(PIP 0)

Global Scriptor Module Communication Sheet

Revision: 4/25/2019

PIP On	(PIP 1)
PIP Position Bottom Center	(PPP 3)
PIP Position Bottom Left	(PPP 5)
PIP Position Bottom Right	(PPP 4)
PIP Position Left Vertical Center	(PPP 0)
PIP Position Right Vertical Center	(PPP 2)
PIP Position Top Center	(PPP 1)
PIP Position Top Left	(PPP 6)
PIP Position Top Right	(PPP 7)
PIP Size Large	(PHS 2)
PIP Size Medium	(PHS 1)
PIP Size Small	(PHS 0)
PIP Swap None	(PPS 1)
Power Off	(PWR 0)
Power On	(PWR 1)
Video Mute Off	(SHU 0)
Video Mute On	(SHU 1)

Appendix B. Update Commands

Aspect Ratio	(SZP?)
Auto Image	(AIG?)
Freeze	(FRZ?)
Input	(SIN+MAIN?)
Lamp Mode	(LPM?)
On Screen Display	(OSD?)
PIP	(PIP?)
PIP Input	(SIN+PIIP?)
PIP Position	(PPP?)
PIP Size	(PHS?)
Power	(PWR?)
Video Mute	(SHU?)